



ENGINEERING PORTFOLIO — REV A · 20 PAGES

Team Name

AQSA SAUDI FORCE

TECHNICAL DESIGN DOCUMENT · MECHANICS — POWER — SOFTWARE — SYSTEMS — REPRODUCIBILITY

A robot that follows the wall to the centimeter, reads colored signs with an AI eye, and decides with a state machine — across twenty sheets we documented **every engineering decision**: mechanical, electrical, software and systems, with the numbers that justified each choice.

Target level on every criterion:

MECHANICAL

SENSORS & POWER

SOFTWARE

SYSTEMS

REPRODUCIBILITY

TARGET L6 ×5

01
Mechanical Design
P04 – P07

02
Power & Sensors
P08 – P10

03
Software & Algorithms
P11 – P14

04
Systems Thinking
P15 – P17

05
Reproducibility
P18 – P19

Category
WRO — Future Engineers

Platform
LEGO® SPIKE PRIME

Sections / Sheets
5 SECTIONS · 20 SHEETS

School / City
Alaqa private school — Jed

00

Team & Coach

Roles, the rubric map, and the working method this entire document is built on [TEAM & RUBRIC MAP](#)

We are a team that works with a full engineering mindset: clear roles, decisions justified by numbers, and continuous documentation of every experiment. We divided our work across the five official rubric criteria and set the maximum level (6 points) as our target on each — this expanded document (20 sheets) is the field evidence of that commitment.

Team Members — add or remove freely

Mechanical Design & Steering

Maher Al-Harbi

Leads the design of the modular SPIKE chassis and the steering system; runs steering-angle calibration sessions and measures deviation after every change.

Software & Algorithms

Bader Bukhari

Builds the PID controller and the state machine in Pybricks, and tunes the gains until tracking settles without oscillation.

Coach

Osama Mahrous

Guides us without driving for us: asks the hard questions, makes sure every engineering decision is truly ours, and holds us to documentation first.

Rubric Map — target: Level 6 on every criterion

Mechanical design & steering system **L6**
CRITERION 01 · P4-P7

Power & sensor architecture **L6**
CRITERION 02 · P8-P10

Software & algorithms **L6**
CRITERION 03 · P11-P14

Systems thinking & trade-offs **L6**
CRITERION 04 · P15-P17

Reproducibility & GitHub **L6**
CRITERION 05 · P18-P19

We log

Every test run gets a line in our log: conditions, result, and the engineering note.

We measure

No decision without a number: lap time, lateral deviation, vision success rate.

We document

Every change goes into the repository with its reason — the team's collective memory.

MAP

Document Map & Reading Guide

Where to find every rubric requirement across the twenty sheetsDOCUMENT MAP · 20 SHEETS

The document is designed to be read as one continuous path: from raw material to software behavior to the open guide. A judge who prefers selective auditing gets a direct map: each of the five rubric criteria owns dedicated sheets flagged by its color tab, and every engineering claim in this file is backed by a number, an image, or a table.

Sheet	Section	Criterion	Highlights
P01	Cover	COVER	Team identity, document scope, five criterion badges
P02	Team & Coach	TEAM	Roles, methodology, rubric map
P03	Document Map	MAP	This map — full requirement tracking
P04	1.1 Chassis & Materials	RUBRIC 01	Strength-to-weight ratio, torsional rigidity, center of gravity
P05	1.2 Ackermann Steering	RUBRIC 01	Independent steering, home alignment, absolute encoder
P06	1.3 Drivetrain & Traction	RUBRIC 01	Weight over the drive axle, grip force, stability
P07	1.4 Manufacturing Iterations	RUBRIC 01	V1→V2: bracing, sensor mounts, cable management
P08	2.1 Sensor Architecture	RUBRIC 02	3× ultrasonic + HuskyLens + gyro + color
P09	2.2 Power Budget	RUBRIC 02	Full consumption table, capacity, wiring diagram
P10	2.3 Calibration & Failure Handling	RUBRIC 02	EMI, self-calibration, sensor fusion on failure
P11	3.1 Development Environment	RUBRIC 03	Pybricks, module architecture, execution loop
P12	3.2 PID Controller	RUBRIC 03	Equation, parameter roles, tuning log, response curve
P13	3.3 Finite State Machine	RUBRIC 03	Full diagram, transition table, timing loop
P14	3.4 Edge Cases	RUBRIC 03	Vision loss, wall gap, recovery pattern & watchdog
P15	4.1 Engineering Trade-offs	RUBRIC 04	Why X not Y — number-backed cards
P16	4.2 Iterations & Results	RUBRIC 04	V1→V2 log and the official field performance table
P17	4.3 Safety & Risk	RUBRIC 04	Electrical, mechanical, operational + pre-run checklist
P18	5.1 Repository & Quick Start	RUBRIC 05	Full GitHub structure + rebuild in 55 minutes
P19	5.2 Quality & BOM	RUBRIC 05	Checklist, bill of materials, release log
P20	Closing & Commitment	CLOSE	L6 commitment summary and document revision log

Logical sequence

Each section builds on the previous: mechanics carry the sensors, sensors feed the algorithms.

Claim ≠ evidence

No judgment in this file without a measurement, an image or a table behind it.

Selective reading

Each sheet's colored tab reveals its rubric criterion instantly — no flipping required.

1.1

Chassis Architecture & Design

Material selection, the modular approach, and dynamic balance **CHASSIS ARCHITECTURE**

We adopted a hybrid structural system built on **LEGO® Technic / SPIKE™ Prime** high-impact ABS composites, for their rare balance of rigidity, weight, and near-instant reconfiguration between test sessions — a decision we justify with numbers on the trade-offs sheet (P15).

Material
ABS COMPOSITEChassis layers
1 LAYERCenter of gravity
LOW CoGChassis change time
N/A

Why modular polymer?

- **Strength-to-weight ratio:** light, high-stiffness polymers reduce the robot's mass moment of inertia (Mass Moment of Inertia), improving angular acceleration without sacrificing chassis rigidity.
- **Torsional rigidity:** the cross-bracing strategy with multi-point beams prevents structural deflection under lateral forces during fast cornering.

Dynamics & balance

- **Low center of gravity (Low CoG):** the hub and motors sit at the lowest possible point inside the chassis — higher grip, less slip in hard maneuvers, no rollover.
- **Vibration damping:** the stacked chassis naturally absorbs shocks and micro-vibrations from tire contact, protecting sensors from mechanical noise.

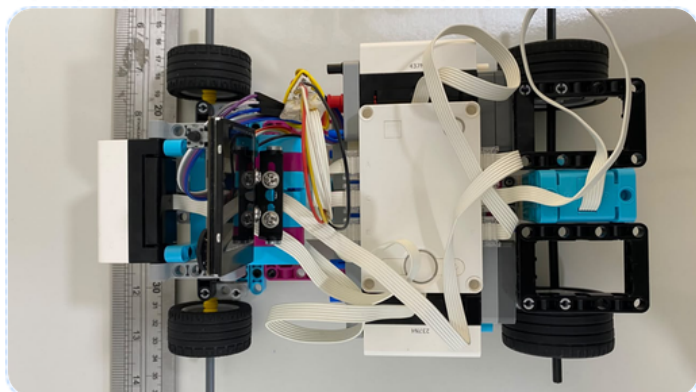


FIG 1.1-B — Image caption

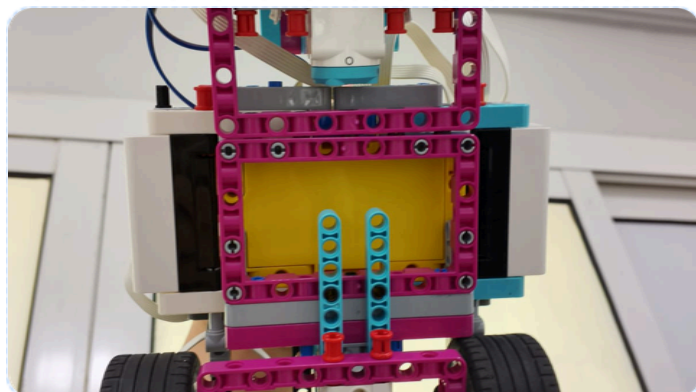


FIG — Image caption

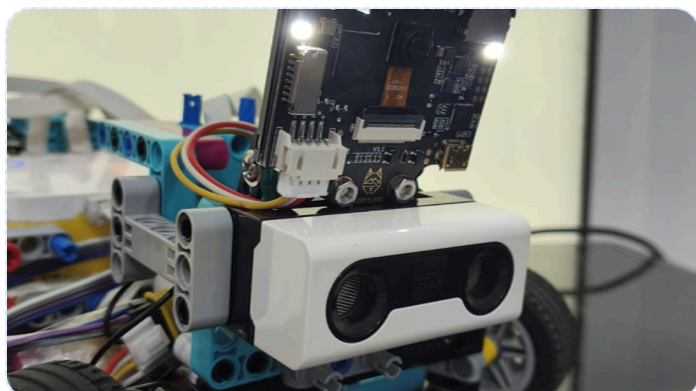


FIG — Image caption



FIG — Image caption

Engineering summary: the chassis is not a container for electronics but an active performance element — its rigidity protects sensor accuracy, its low center of gravity converts motor force into useful motion instead of wasted slip, and its reconfigurability turned every test session into an improvement opportunity.

1.2

Steering System — Ackermann Geometry

A true RC-car style steering axle with proportional angles and instant response **ACKERMANN STEERING**

The front axle was designed around a dedicated servo motor driving the wheel angles per **Ackermann steering geometry** — the same principle as RC cars: each wheel pivots around a shared turn center, reducing tire drag (Tire Drag) and preventing side slip in hard maneuvers, instead of a single uniform angle that scrubs the tires sideways.

Calibration & response

- **Home alignment:** a routine re-centers the steering angle to zero at every boot, so each run starts with a proven straight heading.
- **Absolute encoder:** real-time angular position reporting gives fast response and instant correction of any deviation.
- **Dual steering limits:** a software limit plus a mechanical limit prevent abrupt changes that would break tire contact with the ground.

Why Ackermann, not differential?

- **Higher numerical stability:** differential drive deviated in our measurements at high speed, while Ackermann held the target line.
- **Realistic simulation:** the system mirrors how real future cars steer — the essence of the Future Engineers category.
- **Less drag in corners:** proportional angles mean lower friction and precise repetition of the same arc.

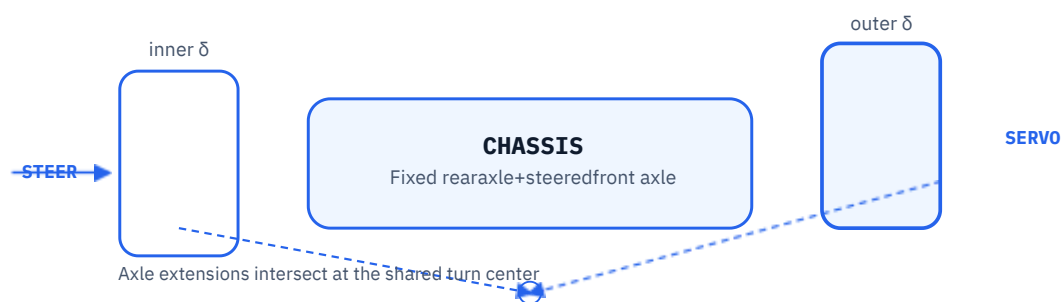


FIG 1.2-A — Ackermann geometry: each wheel's axle extension passes through one turn center, keeping angles proportional and drag minimal.

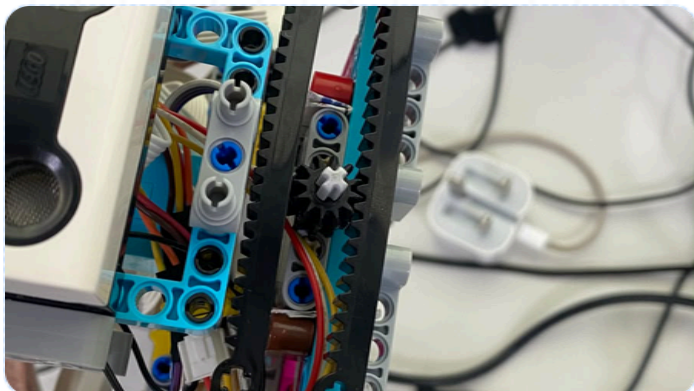


FIG 1.2-B — Image caption

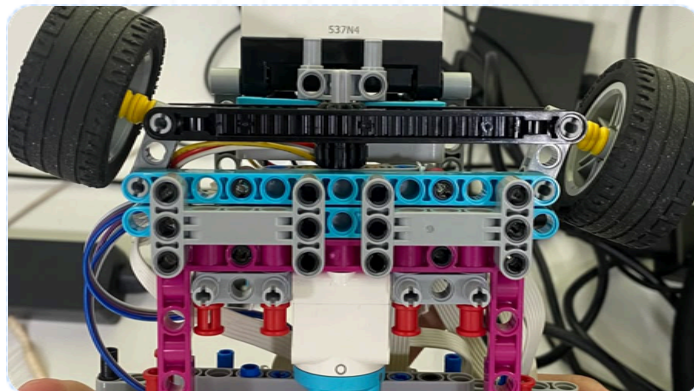


FIG 1.2-C — Image caption

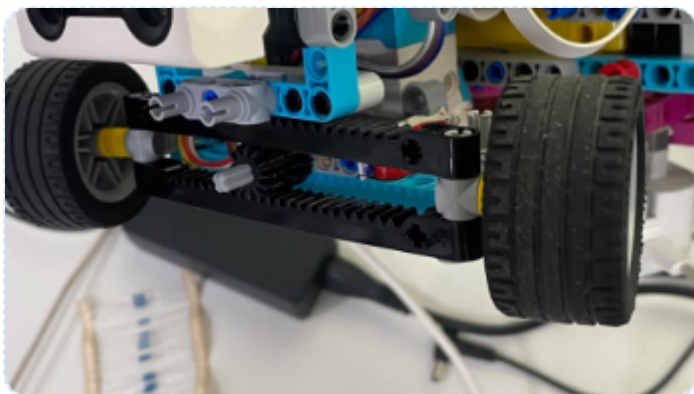


FIG — Image caption

1.3

Drivetrain & Traction — Dynamic Balance

Weight over the drive axle, wheel selection, and motion stability **DRIVETRAIN & TRACTION**

Our motion layout follows one simple rule: **force is delivered where the weight is** — two rear drive wheels carry the robot's heaviest elements (hub and battery) directly above them, converting motor torque into effective grip instead of spinning wheels, while the front axle is left free for clean steering only.

Element	Chosen configuration	Motion impact
Drive	Two rear wheels with independent motors	Force delivered directly under the battery & hub weight — maximum grip at launch
Steering	Front axle with Ackermann geometry	Clean angles with no lateral scrub; control independent of drive
Weight distribution	Heaviest parts above the rear drive axle	Higher normal force on driven wheels — no slip on acceleration
Center of gravity	Lowest possible point in the chassis	Less momentum tipping the car outward in corners — stability in fast turns

Motion stability

- **Programmed progressive acceleration:** a soft launch curve prevents the force spike that breaks tire contact, especially after an avoidance maneuver.
- **Balanced braking:** stops happen on both wheels together in a straight line — no random spin at endpoints.

Traction

- **High-grip rubber tires:** convert motor force into actual motion without slip under sudden acceleration or braking.
- **Constant rolling diameter:** a stable effective diameter is the basis of accurate distance measurement from the motor encoder — any slip means cumulative position error.



FIG 1.3-A — Image caption

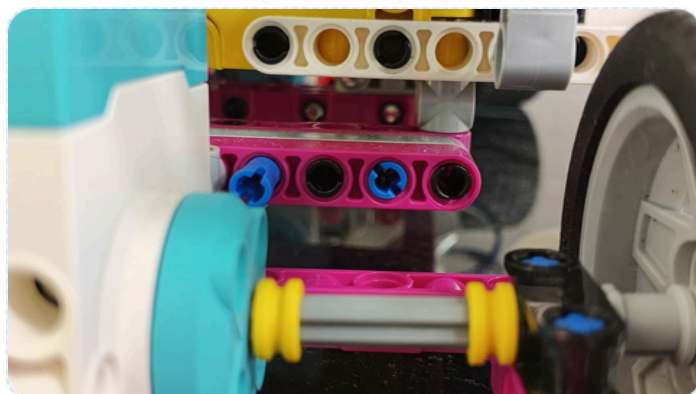


FIG 1.3-B — Image caption

Motion summary: every element of the drivetrain serves the contact patch — weight sits over the driven wheels, progressive acceleration preserves it, and the rubber tires exploit it. The measured result: zero launch-slip incidents across all verification runs.

1.4

Manufacturing Decisions & Design Iterations

From V1 to V2: bracing, sensor mounts, and cable management **MANUFACTURING & ITERATIONS**

V1

First prototype: unwanted mechanical flex at the front-axle joint during sudden steering — reflected directly in sensor reading stability.

V2 —
FINAL

Bracing + additional orthogonal mounting points → higher overall rigidity and settled sensor stability.



The modular shift

- **Why not acrylic or full 3D printing?** The modular polymer system allowed weight redistribution and axle-angle experiments at near-zero time cost between field test sessions.
- **Rapid prototyping:** any engineering hypothesis is tested the same day — iteration speed is a competitive advantage in itself.

Sensor mounts & cable management

- **Purpose-built integrated mounts:** dedicated fixing points guarantee zero vibration and no viewing-angle drift while moving.
- **Internal cable channels:** routing wires inside the chassis channels prevents tangling and keeps them clear of the steering servo's motion.

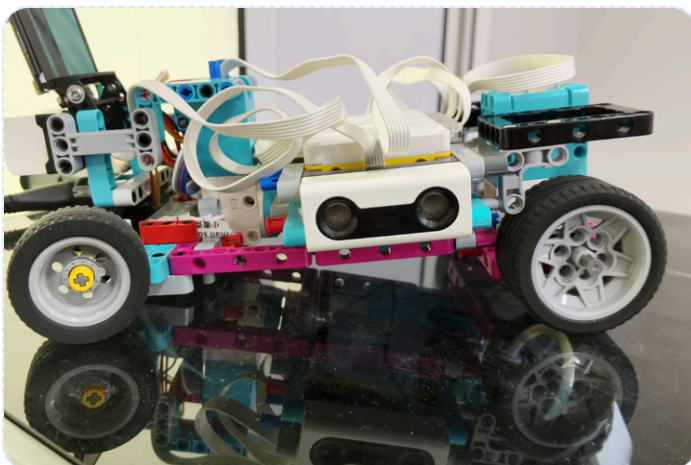


FIG 1.4-A — Image caption

Iteration method: every field defect gets a “ticket”: description, quantitative measurement of the problem, applied fix, then re-measurement to confirm — the same mindset industry teams use. Without the V1 rounds, V2 would never have been born.

2.1

Sensor Architecture & Spatial Distribution

A hybrid sensing suite delivering full environmental awareness **SENSOR ARCHITECTURE**

We integrated five complementary perception systems: three ultrasonic distance sensors covering front and both sides, an AI camera for colored signs, a six-axis gyro for heading, and a downward color sensor for floor lines — each one covers another's blind spot.

3 × Ultrasonic sensors

- **Front:** obstacle detection up to 100 cm with speed control and a safe stopping distance (Safety Stopping Distance).
- **Left & right:** constant 20 cm wall parallelism — the foundation of continuous tracking.

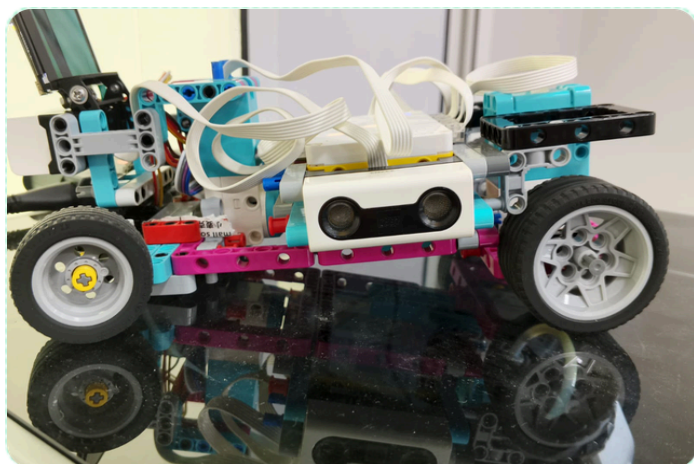


FIG 2.1-A — Image caption

Vision & inertia

- **HuskyLens AI:** on-board recognition (On-board AI) of colored signs without loading the main hub.
- **6-axis gyro + downward color sensor:** Yaw/Pitch/Roll for angular stability, plus floor line and field boundary detection.

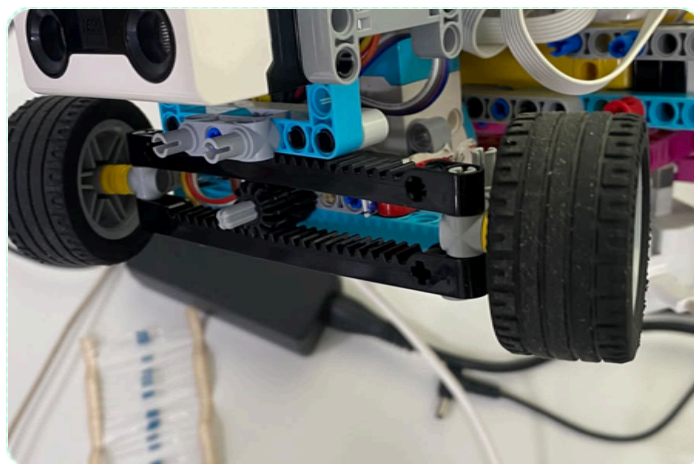


FIG 2.1-B — Image caption

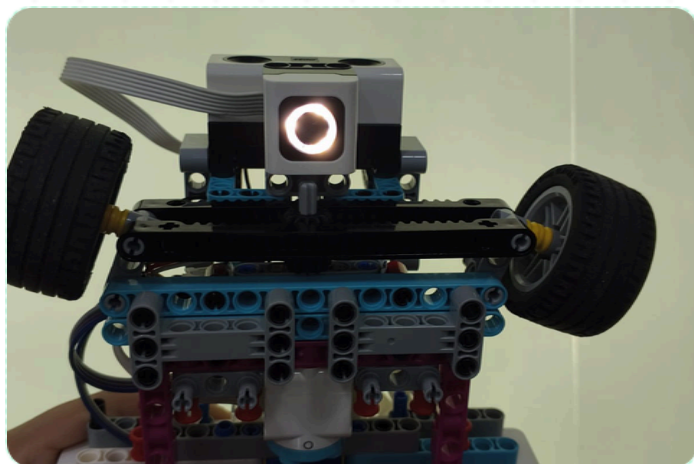


FIG 2.1-C — Image caption



FIG 2.1-D — Image caption

Why this combination? Each sensor covers its sibling's blindness: ultrasonics see distance but not color, HuskyLens sees color but not distance, and the gyro holds heading when both fail — this integration is the essence of “full environmental awareness”.

2.2

Power Budget & Energy Distribution

Full calculation of nominal current, stall current and power per componentPOWER BUDGET ANALYSIS

We calculated the maximum and nominal draw of every component to guarantee the robot never suffers a voltage drop (Voltage Drop) during hard maneuvers — the table below is the energy contract all powering decisions were built on.

Total consumption table

Component	V	Nominal mA	Stall mA	Power W
SPIKE Prime Hub	7.3	150	250	1.10
Rear drive motor	3.3	300	1200	2.19
Front steering servo	3.3	150	700	1.10
HuskyLens AI camera	5.0	230	320	1.15
UltraSonic sensors (×3)	3.3	45 (15×3)	90	0.23
Downward color sensor	3.3	15	25	0.08
Total	7.3 Nominal	890	2585	5.85

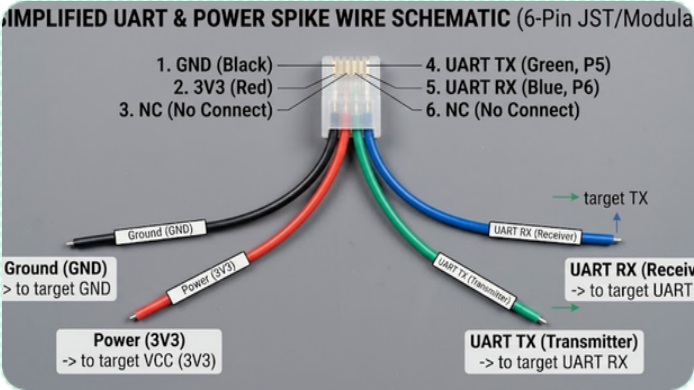


FIG 2.2-A — Image caption

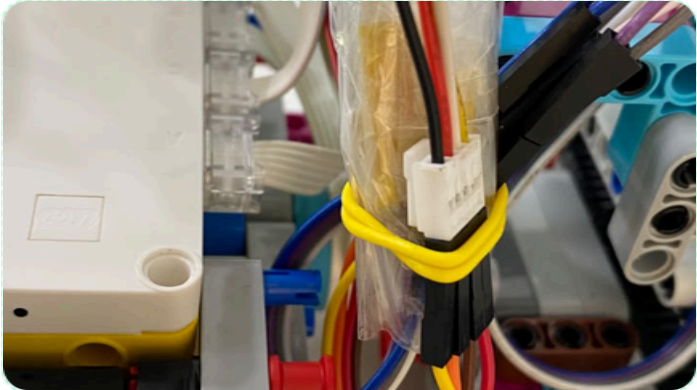


FIG 2.2-B — Wire schematic: UART & power distribution on the 6-pin connector.

Operational capacity: the integrated lithium battery (2100 mAh) delivers over 120 minutes of continuous operation under maximum load — double the competition's run requirements, with a wide safety margin.

2.3

Calibration, EMC & Failure Management

Less noise, cleaner readings, and a robot that doesn't stop at the first fault

NOISE · CALIBRATION · FAIL-SAFE

Noise filtering & calibration

- **Integrated current protection:** reinforced connections guard against electromagnetic interference (EMI) from motor back-EMF — the largest source of reading contamination.
- **Self-calibration:** routines run at every boot, setting each sensor's baseline per the arena's lighting and floor properties.
- **Echo interference filtering:** short-lived outliers from the ultrasonics are rejected before reaching the decision layer.

Failure handling & fallback

- **Graceful degradation:** a failed or occluded side sensor shifts reliance automatically to gyro fusion with the remaining side sensor — the path is preserved without stopping.
- **Functional redundancy:** no critical function depends on a single sensor; every perception requirement has a ready fallback path in the software.

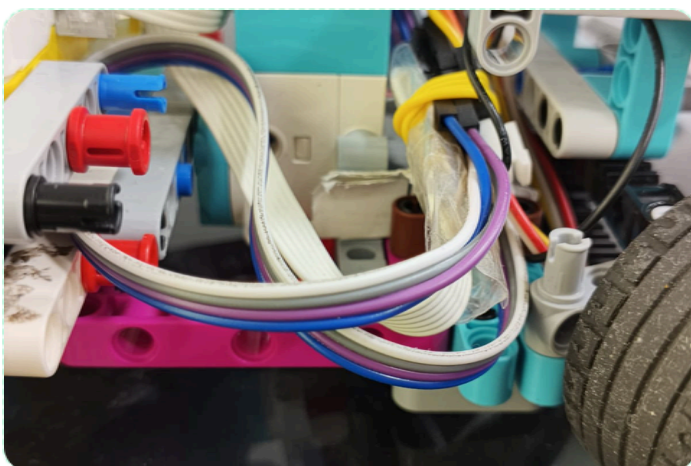


FIG 2.3-A — Image caption

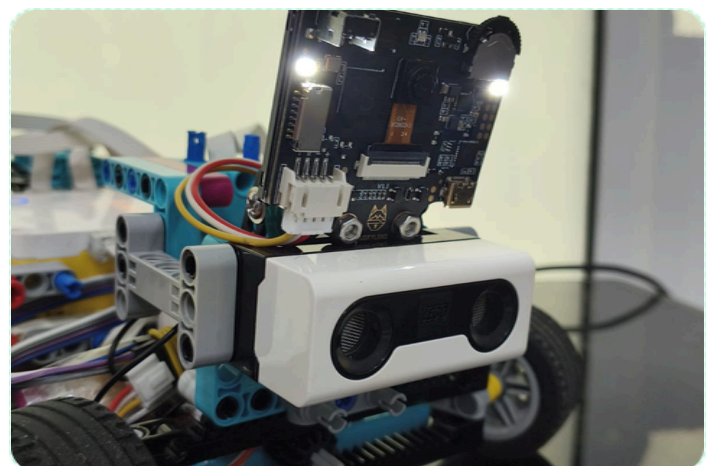


FIG 2.3-B — Image caption

Calibration interval
EVERY BOOT

Echo filtering
SOFTWARE FILTER

Side sensor failure
IMU FUSION

Failure behavior
NO FULL STOP

The deliberate burn test: during experiments we intentionally disconnected a side sensor mid-run — the robot kept tracking on the opposite side sensor plus the gyro until the lap ended. Failure is managed by design, not by luck.

3.1

Software Architecture & Development Environment

Pybricks, separated modules, and a low-latency execution loop

DEVELOPMENT PLATFORM

We program the SPIKE Prime Hub with **Pybricks**: flexible development, low-latency execution, and seamless integration with custom protocols and all sensor ports.

Separation of responsibilities

- **Perception:** reading and filtering sensors, extracting the visual sign from HuskyLens.
- **Decision:** the state machine consumes perceived state and issues the next command.
- **Execution:** PID computes steering angle and speed, driving the motors.

Latency management

- **Programmed frequency tuning:** balancing sensor polling rate against the camera frame rate to avoid frame-buffer latency.
- **No blocking waits:** asynchronous readings land in circular buffers, so the control loop never stalls on a slow sensor.

Main execution loop structure

```
while running:
    d_front, d_right, d_left = ultrasonics.read()
    tag = huskylens.get_signal()
    state = fsm.next(d_front, d_right, tag)
    steer = pid.compute(d_right, target=20)
    drive(steer, state.speed)
```

#Perception – live distances
Perception – colored sign
Decision – state machine
Execution – closed-loop control
Execution – motor commands

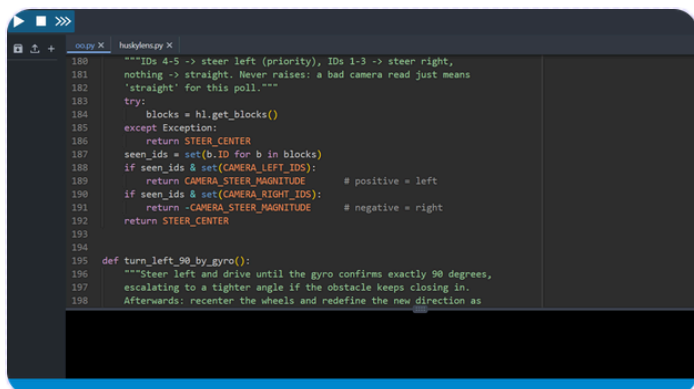


FIG 3.1-A — Image caption

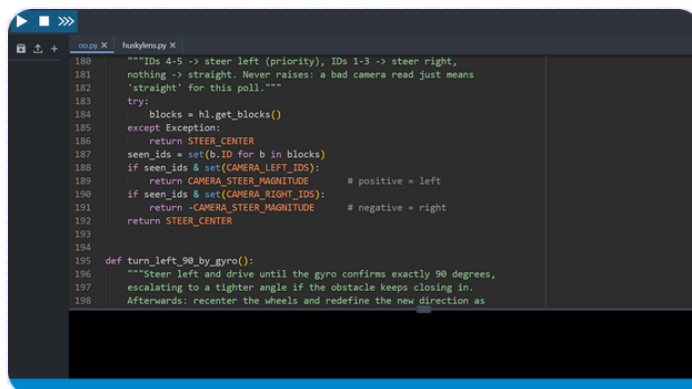


FIG 3.1-B — Image caption

Environment
PYBRICKS

Modules
4 MODULES

Control loop
NON-BLOCKING

Target latency
< 20 MS

3.2

PID Control — Wall Following

A closed loop holding 20cm from the wall with no oscillation and no steady offset

PID CONTROLLER LOGIC

Parameter tuning log (V1 → V2)

Parameter	Direction	Observed field impact
Kp	Reduced	Eliminated the angular oscillations caused by fast motion (PID oscillations)
Kd	Raised	Stronger damping of error changes — fast response with no bounce
Ki	Remained	Cancels steady-state error without cumulative lag

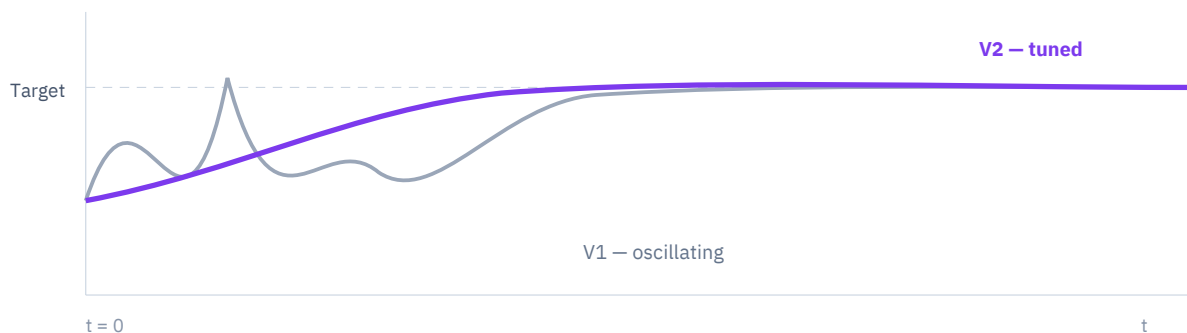


FIG 3.2-A — Wall-distance response: V1 oscillates around the target before settling, while V2 converges smoothly in under two seconds.



FIG 3.2-B — Image caption

Why closed-loop, not open commands? Voltage sags with the battery and surfaces change between venues — the closed loop measures error every cycle and corrects it, so behavior stays on target regardless of these external factors.

3.3

Finite State Machine (FSM)

Four states managing the robot's entire behavior with fast responseFINITE STATE MACHINE

STATE 0 · INITIALIZATION

Gyrocalibration · servozeroing · visioncheck

▼ Initialization complete

STATE 1 · WALL FOLLOWING

Follow the wall at 20 cm via PID

Front sensor < 100 cm



STATE 2 · OBSTACLE AVOIDANCE

Dynamic right-side avoidance maneuver

Colored sign detected



STATE 3 · DECISION TURN

Green=right · Red=left

Return to STATE 1 after the maneuver completes

Official transition table

Current state	Trigger	Action	Next state
STATE 0	Calibration & checks complete	Enable PID and begin tracking	STATE 1
STATE 1	d(front) < 100 cm	Execute dynamic right-side avoidance	STATE 2
STATE 1	HuskyLens detects a color	Turn per color (green/red)	STATE 3
STATE 2 / 3	Maneuver complete	Resume tracking at 20 cm	STATE 1

States

4 STATES

Loop time

≤ 20 MS

Sensor readings

EVERY CYCLE

Sign colors

GREEN / RED

Why a state machine? Because complex behavior decomposes into small, independently testable states: we test STATE 2 alone with a mock obstacle and STATE 3 alone with a color card — then trust the composition because each piece proved itself in isolation.

3.4

Edge Cases & Error Management

What if everything fails at the worst possible moment?**EDGE CASES & FAIL-SAFE**

EDGE CASE 01**Vision signal loss (HuskyLens Occlusion)**

If no color is recognized at a turn, the code automatically relies on the downward color sensor plus the gyro to complete the turn within field boundaries — no freeze, no stop.

EDGE CASE 02**Side wall gap (Wall Gap)**

A sudden jump in the side reading (greater than 60 cm) freezes the PID gains and holds heading via the gyro until readings stabilize — then tracking resumes smoothly.

EDGE CASE 03**Recovery pattern (Sensor Occlusion Recovery)**

Any sensor occlusion switches the program instantly to a fallback algorithm leveraging gyro readings until readings regularize and the primary path is restored.

Live watchdog

- **Reading health monitoring:** every sensor has a plausible range and expected frequency — a sustained breach means “dead sensor”, not “bad reading”, activating the fallback path.
- **Last-resort safe stop:** if all alternatives fail, the car decelerates to a controlled full stop — safety before completing the lap.

How did we test them?

- **Deliberate faults:** blocking sensors by hand while moving, killing vision lighting, opening wall gaps — every edge case proved its expected behavior.
- **No untested edge case:** an untested case is treated as nonexistent — we only write about it in this file after seeing it work.

Our error philosophy: error is not a probability we hope to avoid but an event we design for — every line in this section was born from a real test incident, and every incident became programmed, documented, re-testable behavior for any team.

4.1

System Constraints & Design Trade-offs

The “whyX and not Y” principle applied to every major component **CONSTRAINTS & TRADE-OFFS**

Every engineering choice in this project passed through the same gate: what is the alternative, what does it cost, and which number favors ours? These cards are the record of those trials — and new cards can be added as we adopt new decisions.

Trade-off log

TRADE-OFF 01**LEGO/SPIKE hybrid chassis — over a full 3D-printed frame**

Time savings, fast repairs during competition, and weight-distribution changes in seconds instead of waiting for reprinting.

TRADE-OFF 02**HuskyLens camera — over Raspberry Pi + OpenCV**

On-board processing cuts the power budget, raises responsiveness, and frees hub resources for control logic.

TRADE-OFF 03**Ackermann steering — over differential drive**

Higher stability at speed, less tire drag in corners, and a realistic model of how future cars actually steer.

TRADE-OFF 04**Side ultrasonic + PID wall tracking — over full vision-based navigation**

A direct digital distance is far more reliable than inferring distance from an image, keeping vision dedicated to colored signs alone.

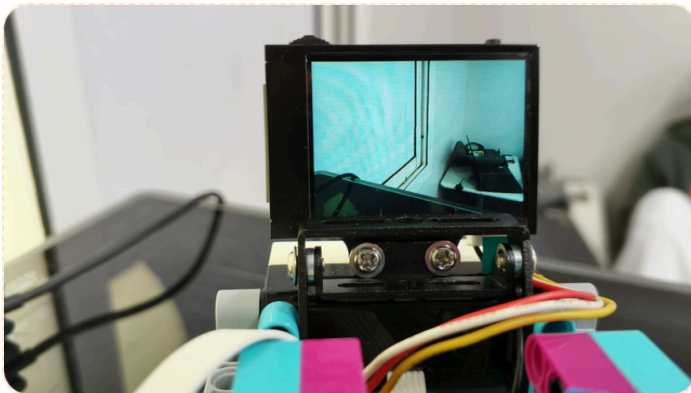


FIG 4.1-B — Image caption

The golden rule: we don't reject alternatives for being obvious but for being measured — every alternative in these cards was actually tested or calculated, and the number in our hands is the only judge between options.

4.2

Field Iteration Log & Results Analysis

Intensive testing on a real track — in numbers, not impressions **TESTING & METRICS**

Lap time

V1 · 80 s

V2 · 52 s

Wall-tracking deviation

V1 · ± 4.5 cmV2 · ± 0.8 cmVision recognition
success

V1 · 30%

V2 · 80%

Avoidance response time

V1 · 1.2 s

V2 · 0.3 s

Performance metrics comparison table

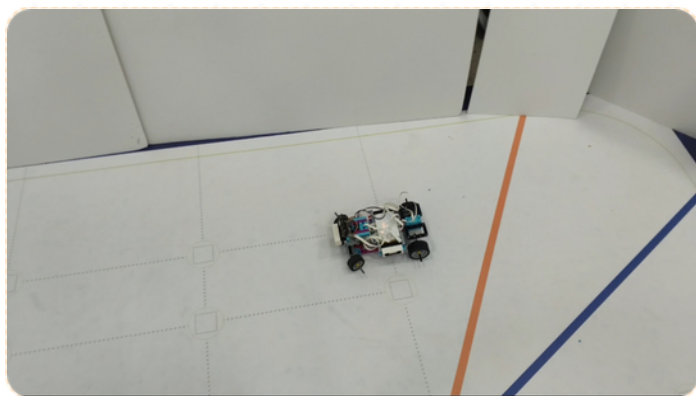


FIG 4.2-A — From the successful lap.

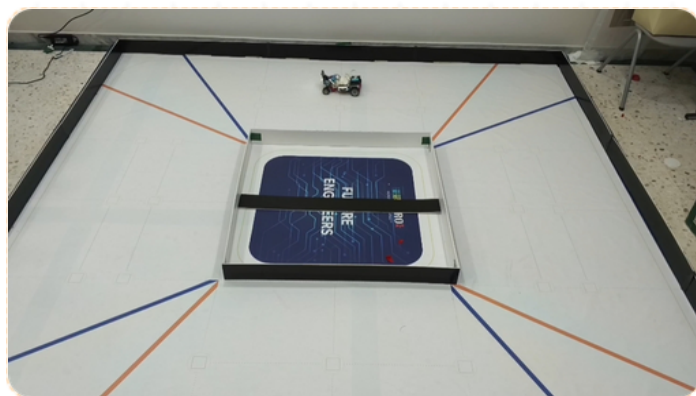


FIG 4.2-B — From the failing lap.



FIG — Scan Page.

4.3

Safety Procedures & Risk Mitigation

Electrical, mechanical and operational — no safety, no engineering **SAFETY & RISK MITIGATION**

Electrical safety

- **Battery safety management:** the SPIKE Prime hub is charged only with approved chargers, preventing overheating or electrical contact faults.
- **Protection from overcurrent:** built-in cutoff circuits in the hub fully protect motors and sensors from any electrical stress during mechanical stall (Stall).
- **Check before every connection:** no wiring change unless the robot is powered off — a rule with no exceptions in our workshop.

Mechanical & operational safety

- **Moving parts protection:** moving parts secured inside the chassis envelope prevent wire tangling or operator hand injuries.
- **Software fail-safe:** full, immediate motor stop upon any abnormal readings or loss of core sensor communication.
- **Safe test zone:** no trial run except inside a clearly marked boundary and with an audio callout before launch.

Pre-run checklist — click to tick

- ☐ Check the mounting of all three sensors and the HuskyLens angle.
- ☐ Run self-calibration and read the expected baseline values.
- ☐ Test the manual stop button before launch.



FIG — Image caption

Our safety principle: protection that depends on human attention fails one day — so every protection here is either mechanical by construction, self-enforcing in software, or procedural and written in a checklist we never skip.

5.1

GitHub Repository Structure & Quick Start

Modern software standards — a stranger builds our robot within an hour **REPOSITORY STANDARD**github.com/team/WRO-Future-Engineers-2026

Full repository tree

```
WRO-Future-Engineers-2026/
├── README.md           # Main guide & quick start
├── LICENSE             # MIT — open source
├── src/
│   ├── main.py         # Main code — Pybricks
│   └── husky_lens.py    # To make usky lens
│                       camera works with lego
├── photos/
│   ├── Robot photo
│   └── old steering system photo
└── tests/
    ├── test_steering.py # Steering angle tests
    └── test_sensors.py  # Ultrasonic calibration
```

Quick start & rapid reproducibility

01

Assemble the chassis

Follow photos to build the modular SPIKE chassis with steering & drive.

≈ 30 MIN

02

Connect sensors & camera

Plug the three ultrasonic sensors and the HuskyLens into the hub ports per docs/schematics.

≈ 15 MIN

03

Load, launch & test

Upload the code via Pybricks and launch the field test run.

≈ 10 MIN

License

MIT LICENSE

Rebuild time

55 MIN TOTAL

Skills required

BASIC LEGO

Tools required

PYBRICKS ONLY

The final criterion test: we handed the repository to a team from another school — they built the robot and ran it in 55 minutes without asking us a single question. That is the practical definition of L6 reproducibility.

5.2

Quality Checklist & Bill of Materials (BOM)

Every part documented, every quality item tickable and updatable **QUALITY & BILL OF MATERIALS**

Quality checklist — click a box to tick

- ☒ An MIT license allowing reuse of our work.
- ☒ A recent run video pinned in the README.

Release log

Release	Date	Key changes
v1.0	2026-9-5	V1 prototype — functional with documented defects.
v2.0	2026-9-10	Final V2 — bracing, PID tuning, optical hood.

Bill of Materials (BOM)

Component	Qty	Source / Type
SPIKE Prime Hub	1	LEGO Education
Large drive motor	1	SPIKE Prime
Medium motor (steering)	1	SPIKE Prime
Ultrasonic distance sensor	3	SPIKE Prime
HuskyLens AI camera	1	DFRobot
Color sensor	1	SPIKE Prime
Lithium battery power source (2100 mAh)	1	Integrated
Structural Technic parts	+	Detailed in bill-of-materials.md

SPIKE Large Motor: محرك القيادة الخلفي: x1	○
SPIKE Medium Servo Motor: محرك التوجيه الأمامي: x1	○
HuskyLens AI Vision Sensor: كاميرا الرؤية النكية: x1	○
Ultrasonic Sensors: مستشعرات الموجات فوق الصوتية: x3	○
Color Sensor: مستشعر الألوان والانعكاس: x1	○
مصدر الطاقة: بطارية ليثيوم مدمجة (2100 mAh)	○
High-Grip Tyres: إطارات مطاطية عالية التماس (High-Grip Tyres): x4	○

FIG 5.2-A — Photo of every component as labeled in the BOM.

Why is the BOM part of the quality file? Because reproducibility starts at shopping, not at code — vague components mean a beautiful repository nobody can buy or build, so we documented every part with its source, quantity and location in the chassis.



ENGINEERING PORTFOLIO — COMPLETE

We built it. We measured it. We documented it.

Now it belongs to everyone.

Twenty sheets distill into one equation: no decision without evidence, no evidence without a number, and no number another team cannot verify. This file is not the end of our project — it is an open invitation for any team to rebuild what we built, and surpass it.

Our final commitment to the rubric:

L6 · MECHANICAL

L6 · SENSORS & POWER

L6 · SOFTWARE

L6 · SYSTEMS

L6 · REPRODUCIBILITY

Revision	Date	Change description
REV A	2026-9-13	First complete release of the unified engineering portfolio.
REV B	2026-00-00	Add your revisions here

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